

# Assessment of the proposed *Metacommunity Ecology* chapter for BCB743

and its relation to a planned chapter on niche and neutral theory

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*Scope. This memo assesses the chapter-contribution presentation metacommunity\_ecology.pptx (Task A2, presentation component) by A. Waglay, B. Effendi, and C. Mc Intyre. It evaluates the proposal against the existing BCB743 material and against a separate, planned chapter on niche and neutral theory, and it records design recommendations for both chapters.*

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## 1. Summary and recommendation

The presentation identifies a real and specific gap, and it proposes a sound device for closing it. BCB743 teaches a multivariate toolbox (ordination, dissimilarity methods, constrained ordination, regression) with considerable rigour, but the ecological theory that gives those tools their purpose is currently scattered across asides and worked examples rather than developed in its own right. The metacommunity framework is a defensible way to organise that theory around the methods already on the syllabus.

Three judgements follow.

- **Accept the proposal in principle, with revision.** The diagnosis on slide 2, namely that the existing tools are not unified under a single framework that explains why communities differ across space, is accurate and aligns with the module's own stated outcomes.
- **Design the two chapters together, not separately.** The proposed metacommunity chapter and the planned niche/neutral chapter describe the same underlying processes at two levels of organisation. Treated as independent units they will overlap and, worse, may present niche and neutral ideas as competing frameworks when the current synthesis treats them as endpoints of one process continuum.
- **Sequence theory first, spatial operationalisation second.** A short niche/neutral chapter should establish the conceptual foundation early; the metacommunity chapter should follow later as the spatial application that connects to variation partitioning, after db-RDA has been taught.

## 2. What the presentation proposes

The deck is a nine-slide conceptual overview. It states the gap (slide 2), defines a metacommunity as a set of local communities connected by dispersal of multiple interacting species (slide 3), lists the four processes that generate community differences (slide 4: environmental filtering, dispersal limitation, species interactions, stochasticity), and presents the four metacommunity paradigms of Leibold et al. (2004) (slide 5: species sorting, mass effects, neutral theory, patch dynamics). It then maps each paradigm onto methods already in the module (slide 6) and sketches a worked example on the South African seaweed data (slide 7: Bray–Curtis, nMDS, db-RDA, variation partitioning). The closing argument (slide 8) is that the chapter would integrate community and spatial ecology, connect niche and neutral theory, and reuse existing module datasets and methods. The deck carries no equations, no R code, and no worked output; it is a proposal sketch rather than a chapter draft.

## 3. Is the gap real?

Yes, and it is more precise than the deck states. The module's published outcomes already commit to this material. The index lists, under the core theoretical framework, "ecological hypotheses about how species assemble across space and time, including niche-based mechanisms, neutral processes, and historical

contingencies”, and one outcome requires students to “know and understand the current hypotheses that explain species assembly processes in space and time (e.g. neutral and niche mechanisms)”. No current chapter develops that theory. The word *metacommunity* appears nowhere in the syllabus.

Where the theory does surface, it does so as an aside to a method. The db-RDA chapter (`constrained_ordination.qmd`) frames the seaweed result as a niche-difference mechanism operating under Beijerinck’s “everything is everywhere, but the environment selects”, which is a species-sorting interpretation in all but name. The *spesim* chapter exposes the generating processes directly, namely a species-abundance distribution (including a neutral zero-sum multinomial sampler), environmental filtering as a Gaussian response on a gradient, and a dispersal kernel. The raw ingredients of assembly theory are therefore present in the course, but they are not assembled. The proposal is a request for that synthesis, and the request is justified.

#### 4. The conceptual relationship: niche, neutral, and the metacommunity paradigms

This is the decisive point for designing both chapters, and it constrains how they should be written.

Niche theory and neutral theory are not opposed accounts of the same phenomenon; they are the endpoints of a continuum of process. Vellend (2010, 2016) reduces community dynamics to four high-level processes, namely *selection* (deterministic fitness differences among species, i.e. niche-based sorting), *drift* (stochastic change in abundances), *dispersal* (movement among localities), and *speciation*. Niche theory is the limit in which selection dominates; neutral theory (Hubbell, 2001) is the limit in which species are assumed ecologically equivalent, so that drift and dispersal alone govern composition. Most real assemblages sit between these limits (Gravel et al., 2006; Chase & Myers, 2011).

The four metacommunity paradigms are positions in that same process space, not a separate theory. Species sorting is selection with moderate dispersal. Mass effects add high dispersal, so that source–sink flux overrides local selection. Patch dynamics turn on colonisation–extinction trade-offs among competitors. The neutral paradigm is drift plus dispersal under species equivalence. Leibold et al. (2004) introduced the four-way typology; Leibold & Chase (2018) and the empirical review of Logue et al. (2011) reframe the same ground in terms of underlying processes rather than discrete archetypes.

The implication for the curriculum is direct. The niche/neutral chapter and the metacommunity chapter are the same content at two altitudes. The first should establish the processes; the second should show how those processes play out across space and how they are tested. If they are written independently, they will duplicate the neutral-theory exposition and risk teaching the four paradigms as rival hypotheses rather than as a cross-classification of selection, drift, and dispersal.

#### 5. Fit with the existing methods curriculum

The method mapping on slide 6 is largely sound. PCA recovers environmental gradients; nMDS and PCoA summarise compositional similarity and  $\beta$ -diversity; cluster analysis groups sites; regression tests environmental drivers. The load-bearing link, however, is variation partitioning, and here the proposal connects to the course more deeply than the students appear to realise.

The standard quantitative test for niche versus neutral signal in a metacommunity is the partition of community variation into an environmental fraction [ $E$ ], a spatial fraction [ $S$ ], their shared component, and a residual (Borcard et al., 1992; Peres-Neto et al., 2006; Cottenie, 2005). That exact analysis already exists in the module. The self-study appendix `two_oceans_appendices.qmd` (Chapter 13b) builds spatial eigenfunctions with `PCNM()`, then calls `vegan::varpart()` to split seaweed  $\beta$ -diversity into thermal and spatial fractions, citing Peres-Neto et al. (2006) and Peres-Neto & Legendre (2010). The lectured db-RDA chapter, by contrast, fits the environmental model only and explicitly sets the spatial coordinates aside.

So the method the proposal needs is taught, but as an unmotivated technical appendix. The metacommunity chapter would give that orphaned analysis its ecological reason for existing, namely separating the environmental (species-sorting) signal from the spatial (dispersal-related) signal. That is a strong argument for adding the chapter, and a clean integration point: the chapter does not introduce a new method so much

as supply the theory that makes an existing one interpretable.

One terminological hazard needs managing. The course already uses the word *partition* in a different sense. Chapter 13a-revised (constrained\_ordination\_v2.qmd) partitions  $\beta$ -diversity *dissimilarity* into turnover and nestedness components (Baselga, 2010), and then into replacement and richness-difference components (Legendre, 2014). That is a decomposition of a dissimilarity coefficient, not the environment–space decomposition of community variation. The two operations share a word and nothing else. The new chapter must keep the two senses distinct or it will confuse students who have just met the first.

## **6. Critical caveats**

A chapter built on this framework must assess its own tools, not advertise them. Three caveats should be written into the chapter rather than left for the lecturer to add.

### **(a) The four-paradigm typology is dated**

The discrete four-way scheme of Leibold et al. (2004) has largely given way to the process-based view (Vellend, 2010; Logue et al., 2011; Leibold & Chase, 2018). The paradigms remain a useful teaching scaffold, but presenting them as the current state of the field would misrepresent it. Teach the typology as history, then move to selection–drift–dispersal as the operative framework.

### **(b) Variation partitioning is a weak test of process**

The [*E*] versus [*S*] partition is suggestive, not diagnostic, and the chapter must say so. Environmental variables are themselves spatially structured, so the shared fraction is typically large and hard to interpret. The pure spatial fraction [*S*] conflates dispersal processes with spatially structured but unmeasured environment, since spatial eigenfunctions absorb autocorrelation from any source (Gilbert & Bennett, 2010; Smith & Lundholm, 2010; Tuomisto & Ruokolainen, 2006; Peres-Neto & Legendre, 2010). Reporting [*S*] as “the neutral component” would be indefensible. The honest claim is that a partition is consistent with a relative balance of processes, under stated assumptions, and requires further testing.

### **(c) The seaweed dataset suits one paradigm, not all four**

The South African seaweed data describe an essentially one-dimensional thermal gradient sampled by contiguous coastal sections. That design is well suited to demonstrating species sorting, which is exactly the conclusion the existing db-RDA chapter reaches under Beijerinck’s law. It is poorly suited to demonstrating mass effects or patch dynamics, which require patchy, dispersal-structured landscapes with replicated patches. The worked example should therefore frame the seaweeds as a species-sorting case study and extend the existing interpretation, not promise a demonstration of all four paradigms it cannot deliver.

## **7. Recommended sequencing and division of labour**

The two chapters should form a conceptual pair, placed at either side of the constrained ordination block.

| Chapter                                           | Content and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Niche & neutral theory ( <i>new, conceptual</i> ) | Selection, drift, dispersal, and speciation as the processes of community assembly (Vellend, 2010); the niche concept and modern coexistence theory (Chase & Leibold, 2003; HilleRisLambers et al., 2012); Hubbell's neutral theory and its testable predictions for the species-abundance distribution, dispersal limitation, and distance decay of similarity (Hubbell, 2001); the niche–neutral continuum. <i>Place early</i> , after the introduction to ordination and $\beta$ -diversity, to motivate why turnover is measured at all. |
| Metacommunity ecology ( <i>the proposal</i> )     | The four paradigms as positions in selection–drift–dispersal space; variation partitioning ( $[E]$ vs $[S]$ ) as the quantitative test and its limitations; a worked seaweed example that reuses the Chapter 13b PCNM and <code>varpart()</code> machinery. <i>Place late</i> , after db-RDA and variation partitioning are in hand.                                                                                                                                                                                                         |

This bookend arrangement closes the standing gap between the module's stated outcomes and its delivered content, and it gives the self-study variation-partitioning appendix a lectured home.

## 8. An integration the proposal misses: *spesim*

The *spesim* package is the controlled setting in which the caveats above become a practical exercise. It already simulates the three processes the chapters turn on, namely a species-abundance distribution (including a neutral zero-sum multinomial), environmental filtering as a Gaussian response on a gradient, and dispersal through a kernel. Students can generate communities under a *known* balance of species sorting and neutral drift, run the  $[E]/[S]$  partition, and check whether it recovers the truth they specified. That exercise demonstrates caveat (b) directly rather than asserting it, and it triangulates with Task A3, the *spesim* package review. The proposal should adopt it.

## 9. The presentation as an assessed artefact (Task A2)

**Strengths.** The gap is correctly diagnosed and tied to the module's own framing. The method mapping is accurate. The chosen dataset reuses material already in the course, which is the right instinct for a teaching chapter. The structure moves cleanly from problem to framework to application.

**Weaknesses relative to a deliverable chapter.** The deck is conceptual only, with no equations, no R, and no output, so the distance to a worked BCB743 chapter is large. The four paradigms are presented without the modern synthesis or the inferential caveats. “Variation partitioning” is invoked on slide 7 without being shown and without being distinguished from the  $\beta$ -diversity dissimilarity partition the students will already have met. The *spesim* opportunity is absent.

**Verdict.** A sound and well-judged proposal. The framework is appropriate, the fit with the module is genuine, and the gap it targets is real. The work that remains is substantial and lies mainly in the worked quantitative example and in the critical framing of what variation partitioning can and cannot establish.

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